Sound Waves

We've seen that if we shake a string under tension, the motion of the string is governed by the wave equation. We've seen several solutions to the wave equation, depending on how our system is set up (very long string, string tied at one end, string tied at both ends). Those solutions apply to any system governed by the wave equation. We'll now look at waves in air (and other materials), and see that we get exactly the same equations. As usual, I won't hold you responsible for the derivation, but you will be responsible for being able to use the results we work out now.

1. Pressure from Moving Windows

To see how pressure turns into sound waves, consider a long rectangular box with a bunch of windows in it that can slide along the box. Think "club sandwich" where each slice of bread is one of our sliding windows. The windows are air-tight, including where they hit the box, so an air molecule that starts between two windows will stay between those two windows forever.

Now we want to look what happens as we slide the windows around. We can start with a finite number of windows, that start out at positions $[x_0, x_1, x_2...]$. We'll assume the windows are evenly spaced with distance dx, so that $x_1 - x_0 = x_2 - x_1 = x_3 - x_2 = dx...$ When we looked at waves on a string, we said that y was the height of the string, which is just the distance the string moved in the vertical direction. If the rectangle is lying flat along the x direction, then the windows are standing upright, and they can only move in the x direction. We'll now call y the *distance* each window has moved along the x-axis, relative to where it started. So $[y_0, y_1, y_2...]$ are the distances out windows have moved, which means their final positions are $[x_0 + y_0, x_1 + y_1...]$.

Before any windows move, the pressure is the same everywhere, and we'll call it P_0. As the windows move around, the pressure between windows will change, and our goal is to figure out how the pressure changes as a result of the motion. If we start with $y_0 = y_1 = y_2... = 0$ then the windows are in their starting position. For a given window, the pressure on either side of a window is still P_0, and there is no *net* force on any of the windows. What happens if $[y_0 = y_1 = y_2... = a]$ for some value a? Now every window has moved over by the same amount (a), so the spacing between windows hasn't changed. If the distance between a pair of windows hasn't changed, then the density of air hasn't changed and so the pressure hasn't changed. Since no pressures have changed, there still can't be any forces. The analogy to the string is that we lift the whole string up by an amount a.

The next case to consider is if y changes linearly; that is say $y_0 = 0$, $y_1 = 0.1$, $y_2 = 0.2...$ More generally, we can say $y_n = \alpha n$ for some constant α. In our example, we would have $\alpha = 0.1$. What happens now? The new position of window n is $x_n + y_n = x_n + \alpha n$. The new position of window $n+1$ is $x_{n+1} + \alpha(n+1)$. The separation between the windows is just the difference of their positions:

$$x_{n+1} + \alpha(n+1) - (x_n + \alpha n) = x_{n+1} - x_n + \alpha(n+1-n) = dx + \alpha \quad (1)$$

The original spacing of the windows was dx, the new spacing is $dx + \alpha$. The windows have all gotten closer together or further apart, but the distance between any pair of windows is the same.

It used to be dx, now it is $dx+\alpha$. The pressure will change, but because every pocket of air changes gets squished by the same amount, the pressure in each pocket changes by the same amount. The *net* force on each window is still zero because the pressure on either side balances.

If we want to draw an analogy to a string, this case of a uniform change in window spacing is like a uniform slope of the string. Tilting a string but keeping the slope uniform makes the tensions pulling on a bit of the string cancel. Similarly, a uniform *change* in the spacing of windows keeps the forces balanced. What we need to get a force is the values of y change their slope. If I am a windows, and one of my neighbors is closer to me than the other, the pressure between me and my close neighor will be higher than between me and my distant neighbor. The pressures do *not* balance, and I will feel a force *away* from the closer neighbor. Just like a string "wants" to straighten itself out, the windows "want" to be evenly spaced.

Finally, we'll let the spacing of the windows be non-uniform. The old volume of an air pocket is Adx where A is the area of the window. The new volume of an air pocket is $A(dx+dy)$ where dy is the slope of y. This gives us equal pressures on both sides of a window if dy is constant, so the *change* in volume between our left and right sides comes from the *change* in dy. The change in volume, and therefore the change in density, comes from the change of dy, but since dy is the slope of y, what we really care about is the *change in the slope* of y. Now we know that the change in density, which is going to give rise to forces, comes from the change in slope of y, just like the *net* force on a string came from the change in slope of y.

2. Equation of State

We have to do a couple last steps now to clean up our work to get to an actual acceleration. This first is to work out how the pressure depends on density. We know a slope of y gives a changing volume, and density ρ_N is N/V for some quantity N (N might be mass, but it might also be number of molecules, or moles of molecules or...). A change in V gives a change in ρ_N. If we now go back to the ideal gas law, we know

$$PV = NkT \tag{2}$$

$$P = \frac{N}{V}kT \tag{3}$$

$$P = \rho_N kT \tag{4}$$

If the temperature is constant, than a 1% increase in density would lead to a 1% increase in pressure. As we change density, we know the pressure stays proportional to density, so we would get a relation

$$\frac{P}{P_0} = \frac{\rho}{\rho_0} \tag{5}$$

This relation between pressure and density is what is called an *equation of state*.

We need the equation of state because as we move our windows around, what we actually calculate is a change in density. What we want for the force, though, is the change in pressure. The equation of state tells us how the pressure changes as we change the density. They can be quite complicated, because the detailed physics of your material can matter, and *how* you change your

density can matter a lot too. If we look at Equation 5, it would be correct if we compressed our air so slowly that the extra heat from compressing it had time to leak out. In practice, sound waves move much faster than heat spreads, so the temperature *does* go up. A much better equation of state when we're talking about sound is

$$\frac{P}{P_0} = \frac{\rho}{\rho_0}^{\gamma} \tag{6}$$

The exact value of γ depends on the gas you're using. A larger γ means the gas is *stiffer* - the pressure goes up more as you compress it. For air, the value of γ is $7/5 = 1.4$ while for helium it is $5/3 = 1.666....$ On a microscopic scale, when you heat up air, some of the energy goes into making the molecules spin while when you heat helium it all goes into making the helium bounce around faster. It takes *more* energy to heat air, so if we add a fixed amount of energy to air, the temperature will go up *less* than for helium.

3. Wave Equation and Speed of Sound

We're ready to put the pieces together now. For a given change in slope of position shift y, the change in pressure goes up if I make the window spacing wider. If I take a pair of windows, the force will be the difference between the pressure from the left side of the left window and the pressure from the right side of the right window. For a given change in slope of position, then a wider window will mean a wider force, because the pressure change is larger across the window. If we take the windows to be massless, the mass we're going to accelerate is the mass of the air in our air pocket. The force will be $A\delta P$ where A is the area of our window, and δP is the change in pressure between windows. If I tell you the slope of the pressure P' then the difference in pressure is $P'\zeta$ where ζ is the space between the two windows. If the windows are massless, then the mass we need to accelerate is just from the air, which is ρV. V is the volume of our chunk of air, which is just $A\zeta$. Putting the mass and the force into Newton's second law we get:

$$F = A\zeta P' \tag{7}$$

$$m = \rho A\zeta \tag{8}$$

$$F = ma \rightarrow A\zeta P' = \rho A\zeta a \tag{9}$$

$$a = \frac{P'}{\rho} \tag{10}$$

Remember that we're worried about changes in position y, so the acceleration is the rate of change of the rate of change of y, just like is was for a string. After a bit more calculus (which I hope you can take on faith), we get that the wave equation for sound is

$$\frac{\partial^2 y}{\partial t^2} = \frac{dP}{d\rho}\frac{\partial^2 y}{\partial x^2} \tag{11}$$

What Equation 11 tells us is that air will have waves in it just like a string. The math is the same, so the solutions will be the same. Just like for a wave on a string, the coefficient on the right is the *square* of the wave speed, so we have the following very fundamental result:

$$c_s^2 = \frac{dP}{d\rho} \tag{12}$$

where c_s is the speed of sound. What Equation 12 is telling us is that if somone tells us how the pressure changes as we change the density, that we can tell you how fast sound waves will move. Equation 12 is very general - it applies to sound waves in *any* medium - liquid, gas, or solid. You might think that you can't compress water, but water does have a speed of sound (about 1500 m/s), so we have an idea from that how much water compresses if we push on it. Stiffer materials mean we have to push harder to get the same change of density, and so the sound speed will go up. That's why air has a relatively low sound speed (around 340 m/s), but steel has one nearly 20 times higher - about 6 km/s[1].

4. Speed of Sound in Air

We can now work out the speed of sound in air. If you look at the equation of state for a gas, then calculus will tell you that $\frac{dP}{d\rho} = \gamma\frac{P}{\rho}$. If you don't believe me, you can say pick a value for P and ρ, then increase ρ by 1% in Equation 6 and see that indeed the ratio of new pressure to new density increased by a factor of 0.01γ. For air, where $\gamma = 1.4$, we then have

$$\frac{c_s^2}{=} 1.4\frac{P}{\rho} \tag{13}$$

Now we go back to the ideal gas law to get $\frac{P}{\rho}$. We know that

$$PV = nkT \tag{14}$$

$$P = \frac{n}{V}kT \tag{15}$$

The ratio $\frac{n}{V}$ is the *number* of things I have per volume, but what I usually measure is the *mass* of air per volume. If the mass of each air molecule is m, then I can multiply the right side by $\frac{m}{m}$ without changing anything, and get:

$$P = \frac{nm}{V}\frac{kT}{m} = \rho\frac{kT}{m} \tag{16}$$

Divide through by a density, and we have

$$\frac{P}{\rho} = \frac{kT}{m} \tag{17}$$

Equation 17 tells us that the *ratio* of the pressure to the density depends only on the thermal energy scale of our gas (kT) and the mass of each air particle. If we plug in Equation 17 into Equation 13 then we can get the speed of sound in air:

$$c_s^2 = \gamma\frac{P}{\rho} = \gamma\frac{kT}{m} \tag{18}$$

$$c_s = \sqrt{\gamma\frac{P}{\rho}} = \sqrt{\frac{\gamma kT}{m}} \tag{19}$$

[1]Maybe this is a part of why Legolas stuck his ear on the ground to hear distant horsement. The sound arrives much faster through the ground. There's a second reason, which we'll see later.

We can calculate the speed of sound with either expression. If we take macroscopic air quantities, where the pressure is $101kPa$ and the density is about $1.3kg/m^3$, then we get a speed of sound in air at standard pressure/temperature of $\sqrt{\frac{1.4\times1.01e5}{1.3}} = 330m/s$. The actual value is 331 m/s, so we're bang on!

For air,

$$c_s = \sqrt{\frac{\gamma kT}{m}} \tag{20}$$

is better at illuminating the physics. The density of air does not appear hear, so the speed of sound should not depend on the density of the air. You probably realize this intuitively if you think about it. The speed of sound at the top of a mountain is the same as the bottom, otherwise a flute would go *horribly* out of tune. You can play the same flute in Denver as you can in Montreal, so that can't happen. It *does* depend on temperature, however. The air molecules are zipping around faster when you heat up the air, so it isn't surprising that sound moves faster too. The speed of sound does depend on the mass of the molecules, so carbon dioxide should have a lower speed of sound, and the gas sulfur hexafluoride should have a *much* lower speed of sound (it does - about 130 m/s).

Let's work out the speed of sound from Equation 20. At room temperature the temperaure is 20C or 293K. The mass of an air molecule (assuming 29 AMU/molecule) will be about $29/(6.02e26)$, and Boltzmann's constant is $1.38e-23$. Put is all together and we have

$$c_s = \sqrt{\frac{1.4 \times 1.38e-23 \times 293}{29/6.02e26}} = \sqrt{117500} = 343m/s \tag{21}$$

The actual speed of sound in air at room temperature? 343 m/s - we are exactly correct! We can also check the speed of sound in helium. Remember, the value of γ changes to $\frac{5}{3}$, and helium doesn't make molecules; it's pure atoms with mass of 4 AMU. That predicts a speed of sound at room temperature of

$$c_{s,He} = \sqrt{\frac{5/3 \times 1.38e-23 \times 293}{4/6.02e26}} = 1007m/s \tag{22}$$

At 20C, wikipedia tells us that the speed of sound in helium is exactly 1007 m/s, so we're once again exactly right.

5. Longitudinal vs. Transverse and Boundary Conditions

If we compare the sound waves in air to the waves on a string, there's one immediately apparent difference. For a string, the motion of bits of the string is perpendicular to the direction of the wave. If I hold a string horizontal and send a wave down it, the string moves vertically. These are called *transverse* waves, because the particle/string motion is transverse to the wave motion. For air, though, the motion of the air is *parallel* to the direction of the wave. The are called *longitudinal* waves, because the particle/string motion is along the direction the wave moves. These are likely less familiar to you, but since the math is the same, the behavior is the same. *If you've seen one wave, you've seen them all.*

We can also look at how the boundary conditions we've seen translate from strings to air. For a string, y is the height the string moves. For air, in the notation we've used y is the distance along the wave a bit of air moves, relative to where it started. What happens at the end of a closed pipe? Well, we can't push air through the closed end of the pipe, so y can't go negative. If y were to go positive, that means that the air right next to the end pulls away. In turn, that means there's zero air in that little space and so we get a vacuum and a very large pressure change. As we've seen, the actual pressure changes for sound waves (at reasonable volumes) is tiny compared to atmospheric pressure, so we can't actually peel the air next to the pipe end away from it[2]. If y can't be negative, and y can't be positive, then y must equal zero. A closed pipe end then behaves exactly like a string nailed down at the end, because $y = 0$ in both cases.

Just like we can't have a kink in a string, we can't have a jump in pressure. If the pressure jumps by a finite amount at a boundary, then the force on a given area is finite and set by the pressure jump. That force operates on a volume around the jump, but if the pressure jumps we can make that space as narrow as we like. As the space becomes arbitrarily narrow, while the force stays constant, the acceleration goes to infinity. When that happens, the air moves virtually instantly to smooth out the pressure jump, so we know that the pressure is always continuous[3].

At the end of an open pipe, our air hits the atmosphere, and so the pressure has to match atmospheric. We know that if y is constant then we have the same pressure we started with, because the slope of y is how we change pressure. If the pressure at the end has to match atmospheric, then the slope of y has to be zero. Otherwise we would have a jump in pressure. This is exactly the same as a loose string, where the slope of y had to be zero, or y had to be flat at the untied end of a string. An open pipe behaves exactly like a loose string.

6. Summary and Discussion

We've seen that if we push on air, we get sound waves in exactly the same way that we get waves on a string. Just like the wave speed in a string is $\sqrt{\frac{T}{\lambda_m}}$, the wave speed in air, or the sound speed, is $\sqrt{\frac{dP}{d\rho}}$ where $\frac{dP}{d\rho}$ is how much the pressure changes as we compress the medium. For the particular case of gasses, $\frac{dP}{d\rho} = \gamma\frac{P}{\rho}$ where the adiabatic index γ is a dimensionless number that describes the stiffness of the gas. A larger γ means the air heats up more as you compress it[4]. For air, γ is

[2]This is not actually true in situations where we're putting a lot of power into a system. The best-known example (at least that I know of) is ship/submarine propellers. If you try to spin a propeller too fast, the water does actually lose contact with the propeller and you get vacuum. Eventually the water comes crashing back, and you get very loud bangs as the vacuum collapses. This is called *cavitation* and sets a practical limit on how fast a submarine can spin its propeller while trying to remain quiet.

[3]This is true for anything we'll look at. If the speed of air moving becomes larger than the speed of sound, then this rule breaks down because the air won't move any faster than the speed of sound. These jumps in pressure are called *shock waves*. As you might expect, a jump in pressure makes a very large sound, and is a huge part of the reason supersonic flight never really took off commercially.

[4]Techincally, if you have $E = \alpha kT$ where E is the average energy of a molecule, then $\gamma = 1 + \frac{1}{\alpha}$. If you did the bonus problem on PS1 you would have seen this for the case of $\alpha = 3/2$, the standard ideal gas case. For air, there

very close to 1.4, while for atomic gasses like helium, argon, neon... then $\gamma = 5/3 = 1.666....$ As temperatures go up, you can find more ways to put energy into a molecule, so γ decreases, but for the regimes we'll worry about, you can assume γ is a constant[5].

We could also use the ideal gas law to rewrite the sound speed as $\sqrt{\gamma \frac{kT}{m}}$ where m is the average mass of an air molecule. If you remember the one bit of statistical physics I invoked, then the average energy of an air molecule is $\frac{1}{2}kT$ in each direction, so the average speed of an air molecule in one direction is

$$\frac{1}{2}mv^2 = \frac{1}{2}kT \tag{23}$$

$$v = \sqrt{\frac{kT}{m}} \tag{24}$$

For typical conditions, the speed of sound is related to, but slightly *faster* than the speed the individual air molecules are moving.

Finally, we saw that the boundary conditions for a sound wave in a pipe map directly onto the boundary conditions for a wave on a string. If the pipe is closed at one end, the air can't move and so it behaves just like a string nailed down. If the pipe is open, then the pressure has to match atmospheric pressure, and the textitslope of the displacement y has to equal zero.

A wave on a string looks just like a sound wave in air. If you've seen one wave, you've seen them all!

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are two extra ways to put energy into a molecule because you can spin the molecule in two directions, and $\alpha = 5/2$. The quantization of angular momentum is what prevents you from getting the third direction spinning. Had you lived in the 1800's, then just by measuring the speed of sound you could have figured out something was broken with classical physics!

[5] At 1000C, γ for air drops to 1.365, which is not very different from 1.4.